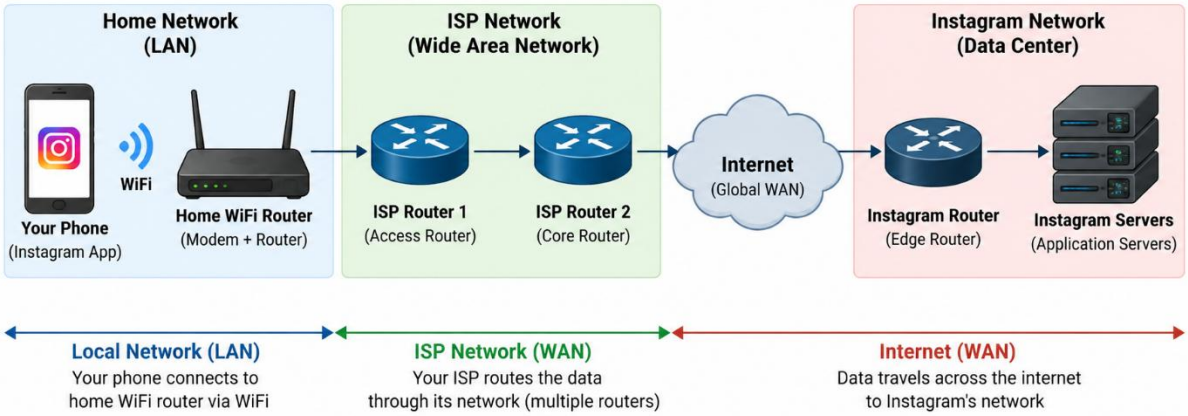
Wide Area Networks

1.Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



Explanation

1. Phone: The Instagram app sends a request to access posts, reels, or messages.
2. Home Wi-Fi router: Connects the phone to the internet and typically translates its private IP address to a public IP address using NAT.
3. ISP Router 1 (Access Router): Connects the home network to the Internet Service Provider (ISP).
4. ISP Router 2 (Core Router): Forwards data through the ISP's wider network. These are two example WAN devices.
5. Internet (WAN): Carries the data across interconnected networks toward Instagram.
6. Instagram Router: Represents the network's edge routing traffic toward the service infrastructure.
7. Instagram Servers: Process the request and send the requested content back to the phone.

2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Speed	Cost	Real-Life Usage
MPLS	High and reliable	High	Connecting multiple corporate offices, such as a Zomato office network.
Leased Line	Dedicated and consistent	High	Connecting critical business locations or data centers, such as ticketing infrastructure.
Broadband Internet	Varies by plan and network congestion	Low to moderate	Home internet for browsing, streaming, and using apps like Instagram.

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

A VPN (Virtual Private Network) creates an encrypted, private tunnel over a public WAN, such as the internet, to protect your data while using public Wi-Fi at a coffee shop.

When you open Spotify or YouTube, your device sends its internet traffic through the VPN tunnel to a VPN server. The VPN encrypts the traffic between your device and the VPN server, helping prevent others on the public Wi-Fi from reading your data. The VPN server then forwards your request to Spotify or YouTube and returns the response through the tunnel.

Example: Even when connected to a coffee shop's public Wi-Fi, a VPN helps protect your internet traffic from local eavesdropping. However, it does not guarantee complete anonymity or protect against every online threat.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Recommended WAN Technology: MPLS (Multiprotocol Label Switching)

Two reasons:

1. Reliable connectivity: MPLS provides reliable, predictable network performance for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi, supporting inventory updates and order processing.

2. Security and traffic management: MPLS supports private network connectivity and traffic prioritization, helping warehouses exchange business data efficiently and prioritize critical applications.